

Problem

A transducer is modeled with a current source I_s and a parallel resistance R_s . The current at the terminals of the source is measured to be 9.975 mA when an ammeter with an internal resistance of $20\ \Omega$ is used.

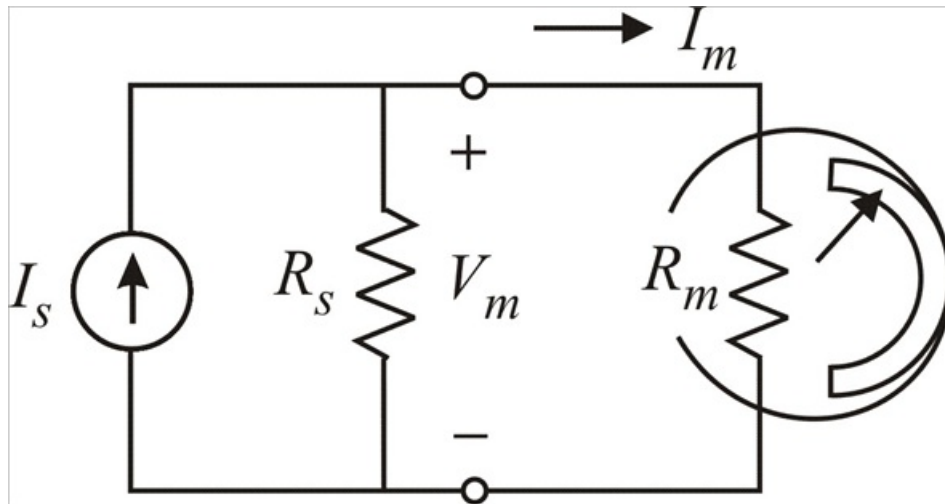
(a) If adding a 2-k Ω resistor across the source terminals causes the ammeter reading to fall to 9.876 mA, calculate I_s and R_s .

(b) What will the ammeter reading be if the resistance between the source terminals is changed to 4 k Ω ?

Step-by-step solution

Step 1 of 7

Circuit diagram from the given data is,



Step 2 of 7

$$I_m = 9.975\text{ mA}$$

$$R_m = 20\ \Omega$$

$$V_m = I_m R_m$$

$$V_m = 9.975 \times 10^{-3} \times 20$$

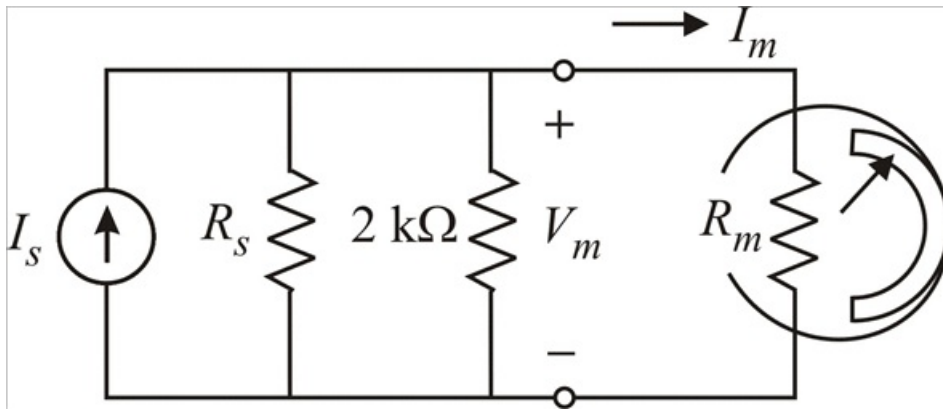
$$V_m = 0.1995\text{ V}$$

$$I_s = \frac{V_m}{R_s} + I_m$$

$$I_s = \frac{0.1995}{R_s} + 9.975\text{ m} \quad \dots\dots\dots(1)$$

Step 3 of 7

(a) Adding a $2\text{k}\Omega$ resistor across the source terminals,



Step 4 of 7

$$I_m = 9.876 \text{ mA}$$

$$R_m = 20\Omega$$

$$V_m = I_m R_m$$

$$V_m = 9.876 \times 10^{-3} \times 20$$

$$V_m = 0.19752 \text{ V}$$

$$I_s = \frac{V_m}{R_s} + \frac{V_m}{2\text{k}} + I_m$$

$$I_s = \frac{0.19752}{R_s} + \frac{0.19752}{2\text{k}} + 9.876 \text{ m}$$

$$I_s = \frac{0.19752}{R_s} + 9.97476 \text{ m} \quad \dots\dots\dots(2)$$

Step 5 of 7

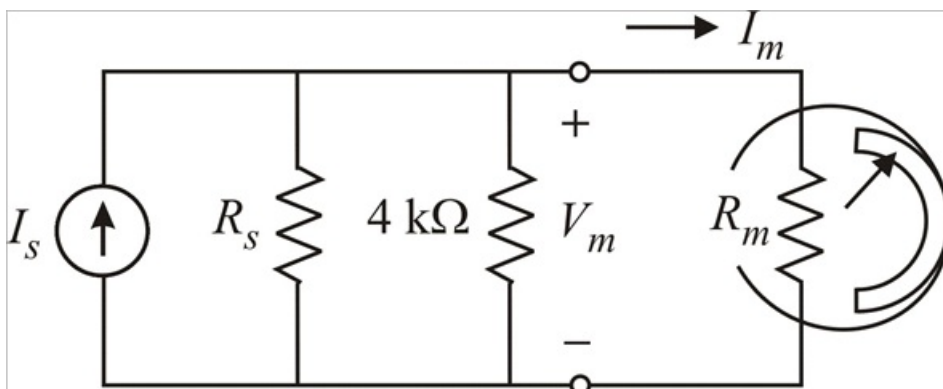
By trial and error method,

$$R_s = 8 \text{ k}\Omega$$

$$I_s = 10 \text{ mA}$$

Step 6 of 7

(b)



$$V_m = I_m R_m$$

$$V_m = 20I_m$$

$$I_s = \frac{V_m}{R_s} + \frac{V_m}{4k} + I_m$$

$$10\text{m} = \frac{20I_m}{8k} + \frac{20I_m}{4k} + I_m$$

$$10\text{m} = (2.5\text{m})I_m + (5\text{m})I_m + I_m$$

$$10\text{m} = (1.0075)I_m$$

$$I_m = \frac{10\text{m}}{1.0075}$$

$$\boxed{I_m = 9.926\text{mA}}$$